1) $A=\{1,2,4,6,\{1,2\}\}$, $B=\{\{1\},\{2\},4,3\}$

a) $A\cup B = R$

$R=\{1,2,4,6,\{1,2\},\{1\},\{2\},3\}$

b) $A\cap B = R$

$R=\{4\}$

c) $A-B=R$

$R=\{1,2,6,\{1,2\}\}$

d) $B-A=R$

$R=\{\{1\},\{2\},3\}$

e) $P(B)=\{\{\},\{\{1\}\},\{\{2\}\},\{4\},\{3\},\{\{1\},\{2\}\},\{\{1\},4\},\{\{1\},3\},$
$\{\{2\},4\},\{\{2\},3\},\{4,3\},\{\{1\},\{2\},4\},\{\{1\},\{2\},3\},\{\{2\},4,3\}$
$\{\{1\},\{2\},4,3\},\{\{1\},4,3\}$

2) $A=\{0,1,2\}$, $B=\{x,y\}$ e $C=\{\#\}$

a) $A\times B=R$

$R=\{(0,x),(0,y),(1,x),(1,y),(2,x),(2,y)\}$

b) $(A\times B)\times(C\times B)$

$(A\times B)=\{(0,x),(0,y),(1,x),(1,y),(2,x),(2,y)\}$

$(C\times B)=\{(\#,x),(\#,y)\}$

$(A\times B)\times(C\times B)=\{((0,x),(\#,x)),$
$((0,x),(\#,y)),$
$((0,y),(\#,x)),$
$((0,y),(\#,y)),$
$((1,x),(\#,x)),$
$((1,x),(\#,y)),$
$((1,y),(\#,x)),$
$((1,y),(\#,y)),$
$((2,x),(\#,x)),$
$((2,x),(\#,y)),$
$((2,y),(\#,x)),$
$((2,y),(\#,y))\}$

c) $P(B) = \{\{\}, \{x\}, \{y\}, \{x, y\}\}$

D) $P(B \times C)$

$B \times C = \{(x, \#), (y, \#)\}$

$P(B \times C) = \{\{\}, \{(x, \#)\}, \{(y, \#)\}, \{(x, \#), (y, \#)\}\}$

e) $A \times (B \times C) =$

$A = \{0, 1, 2\}$

$B \times C = \{(x, \#), (y, \#)\}$

$A \times (B \times C) = \{$
$(0, (x, \#)),$
$(0, (y, \#)),$
$(1, (x, \#)),$
$(1, (y, \#)),$
$(2, (x, \#)),$
$(2, (y, \#))$
$\}$

f) $P(A \times (B \times C)) \to 2^6 \to 64$ subconjuntos

$A \times (B \times C)$

= todas
≠ simétrica
< > transitiva
≤ ≥ Reflexiva e transitiva

3)

a) $a = b$, sobre $N \times N \to$ simétrica, transitiva e Reflexiva

b) $a \neq b$, sobre $N \times N \Rightarrow$ simétrica

c) $a \leq b$, sobre $N \times N \Rightarrow$ Reflexiva e transitiva

d) $a > b$, sobre $N \times N \Rightarrow$ transitiva

4)

a) 7

b) $X = \{1, 2, 3, 4, 5\}$

c) 6

d) $X \times Y \to \{$
$(1,6), (1,7), (1,8), (1,9), (1,10),$
$(2,6), (2,7), (2,8), (2,9), (2,10),$
$(3,6), (3,7), (3,8), (3,9), (3,10),$
$(4,6), (4,7), (4,8), (4,9), (4,10),$
$(5,6), (5,7), (5,8), (5,9), (5,10)$
$\}$

e) 8

5)

a) Passo Base = Verdadeiro

lado esquerdo
$1^3 = 1$

lado direito
$(1)^2 = 1$

Hipótese de indução
Suponha que seja verdade p/ n:
$$1^3 + 2^3 + \dots + n^3 = (1 + 2 + \dots + n)^2$$

Queremos provar:
$$1^3 + \dots + n^3 + (n+1)^3 = (1 + 2 + \dots + n + (n+1))^2$$

Usando a Hipótese:
$$= (1 + 2 + \dots + n)^2 + (n+1)^3$$

Sabemos que:
$$1 + 2 + \dots + n = \frac{n(n+1)}{2}$$

substituindo:
$$= \left(\frac{n(n+1)}{2}\right)^2 + (n+1)^3$$

colocando $(n+1)^2$ em evidência
$$= (n+1)^2 \left(\frac{n^2}{4} + (n+1)\right)$$
$$= (n+1)^2 \left(\frac{n^2 + 4n + 4}{4}\right)$$
$$= (n+1)^2 \left(\frac{(n+2)^2}{4}\right)$$
$$= \left(\frac{(n+1)(n+2)}{2}\right)^2$$

que é:
$$(1 + 2 + \dots + (n+1))^2$$

D S T Q Q S S

b)

Passo Base ⇒ verdadeiro

$n=1 \longrightarrow 1=1^2$

Hipótese

$1+3+\dots+(2n-1)=n^2$

Adicionar próximo ímpar:

$(2n+1)$

$=n^2+(2n+1)$

$=(n+1)^2$

c) Passo Base ⇒ verdadeiro

$n=4 \longrightarrow 4!=24$

$2^4=16$

Hipótese

$n! \geq 2^n$

Provar:

$(n+1)! \geq 2^{n+1}$

Sabemos:

$(n+1)! = (n+1)n!$

usando hipótese:

$\geq (n+1)2^n$

Como $n \geq 4$ então $(n+1) \geq 2$

Logo:

$\geq 2 \cdot 2^n$

$= 2^{n+1}$